

but are still unsure of how to write programs on their own. They may feel that programming hasn't "clicked" for them. They may be able to solve the practice exercises from their tutorials but still struggle to picture what a complete program "looks like." By first copying and then later re-creating the games in this book, they'll be exposed to how the programming concepts they've learned are assembled into a variety of real programs.

The people in the second group are those who are new to programming but are excited and a bit adventurous. They want to dive right in and get started making games, simulations, and number-crunching programs right away. They're fine with copying the code and learning along the way. Or perhaps they already know how to program in another language but are new to Python. While it's no substitute for a complete introductory Python course, this book contains a brief introduction to Python basics and how to use the debugger to examine the inner workings of a program as it runs.

Experienced programmers might have fun with the programs in this book as well, but keep in mind that this book was written for beginners.

About This Book

While the bulk of this book is dedicated to the featured programs, there are also extra resources with general programming and Python information. Here's what's contained in this book:

Projects The 81 projects are too numerous to list here, but each one is a self-contained mini-chapter that includes the project's name, a description, a sample run of the program's output, and the source code of the program. There are also suggestions for experimental edits you can make to the code to customize the program.

Appendix A: Tag Index Lists all of the projects categorized by their project tags.

Appendix B: Character Map A list of character codes for symbols such as hearts, lines, arrows, and blocks that your programs can print.

How to Learn from the Programs in This Book

This book doesn't teach Python or programming concepts like a traditional tutorial. It has a learn-by-doing approach, where you're encouraged to manually copy the programs, play with them, and inspect their inner workings by running them under a debugger.

The point of this book isn't to give a detailed explanation of programming language syntax, but to show solid examples of programs that perform an actual activity, whether it's a card game, an animation, or exploration of a mathematical puzzle. As such, I recommend the following steps:

1. Download the program and run it to see what the program does for yourself.